De La Salle-College of Saint Benilde
DSTALGO | Research Assignment

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Data Structures and Algorithms (DSA) are important concepts in computer science and programming. Data structures are ways of organizing and storing data to allow efficient access and management. Algorithms are step-by-step procedures used to solve problems or perform specific tasks. Together, they help programmers create efficient and well-organized programs.

1. **Linear Data Structure**

   A linear data structure is a type of data structure in which elements are arranged in a sequential order, meaning they are placed one after another. Each element has a specific position within the sequence. Common examples of linear data structures include arrays, linked lists, stacks, and queues.

   - **One-Dimensional Array**

     A one-dimensional array is a linear data structure that stores multiple elements of the same data type in a single sequence. Each element is identified by an index, which is used to access the element. Array indexes commonly start at 0, meaning the first element is located at index 0. One-dimensional arrays are useful for storing and accessing a collection of related values.

2. **Big O Notation**

   Big O notation is a method used to describe the efficiency or complexity of an algorithm as the size of the input increases. It is commonly used to describe the amount of time or memory an algorithm may require. Big O notation helps programmers understand and compare the performance of different algorithms. Common examples include O(1), O(log n), O(n), and $O(n^2)$.

3. **Basic Searching Algorithms**

   Searching algorithms are techniques used to locate a specific item or value within a collection of data. The appropriate searching algorithm may depend on how the data is organized, such as whether the data is sorted or unsorted. Two basic searching algorithms are Linear Search and Binary Search.

- **Linear Search**

  Linear Search is a basic searching technique that checks elements one by one, starting from the beginning of the array. The process continues until the desired value is found or all elements have been examined. Linear Search does not require the data to be sorted, making it useful for simple or unsorted data. Its worst-case time complexity is O(n).

- **Binary Search**

  Binary Search is a more efficient searching technique that is used when the data is sorted. Instead of checking every element, it checks the middle element and determines whether to continue searching in the left or right half. This process is repeated by dividing the remaining search area in half until the desired value is found or there are no elements left to check. Binary Search has a time complexity of O(log n), making it efficient for searching large, sorted collections.

References

GeeksforGeeks. (2026, March 28). *Big O notation*. https://www.geeksforgeeks.org/dsa/analysis-algorithms-big-o-analysis/

GeeksforGeeks. (2026, February 3). *Introduction to linear data structures*. https://www.geeksforgeeks.org/dsa/introduction-to-linear-data-structures/

GeeksforGeeks. (2026, March 3). *Searching algorithms*. https://www.geeksforgeeks.org/dsa/searching-algorithms/